# Spatial Incidence, Skew Lines, Normals, and Orthogonal Projection

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In a plane, two different nonparallel lines intersect. In three dimensions they may instead be skew, so a picture alone is no longer a reliable test for intersection. The elementary incidence rules become clearer when lines and planes are described as affine subspaces, as in Strang's MIT 18.06 linear-algebra lectures.[1]

## §1.1 Lines and planes are affine spans

Given distinct points $p, q \in \mathbb{R}^3$, their line is

$$L = p + \mathbb{R}(q - p).$$

Given $p$ and two linearly independent directions $u, v$, the plane they span is

$$\Pi = p + \operatorname{span}\{u, v\}.$$

These formulas prove several incidence statements at once. Three noncollinear points determine a unique affine plane; a line and a point not on it determine a unique plane; two intersecting lines with different directions determine a unique plane.

When directions are independent, uniqueness follows because any plane containing the supplied points must contain their difference vectors and therefore their span. Dependence is the exceptional case: three collinear points lie on infinitely many planes.

## §1.2 Skew lines are detected by a scalar triple product

Consider two lines

$$L_1 = p + tu, \qquad L_2 = q + sv, \qquad u, v \neq 0.$$

If $u$ and $v$ are linearly dependent, the lines are parallel or coincident. Suppose instead $u \times v \neq 0$. The two lines lie in a common plane exactly when the displacement $q - p$ is in $\operatorname{span}\{u, v\}$, equivalently

$$\boxed{(q - p) \cdot (u \times v) = 0.}$$

If that scalar triple product vanishes, the nonparallel lines are coplanar and therefore intersect. Otherwise they are skew.

[1] https://web.mit.edu/18.06/www/index.shtml

**Example 1.2.1**

Take $L_1 = (t, 0, 0)$ and $L_2 = (0, s, 1)$. Here $u = (1, 0, 0)$, $v = (0, 1, 0)$, and $q - p = (0, 0, 1)$. Thus

$$(q - p) \cdot (u \times v) = (0, 0, 1) \cdot (0, 0, 1) = 1,$$

so the lines are skew even though their horizontal projections intersect.

A unit normal to both direction vectors is $n = (u \times v)/\|u \times v\|$. The shortest distance between nonparallel lines is

$$d(L_1, L_2) = \frac{|(q - p) \cdot (u \times v)|}{\|u \times v\|}.$$

To derive it, decompose the displacement $q - p$ into a component in the plane spanned by $u, v$ and a perpendicular component along $n$. Varying $s, t$ cancels the in-plane component, but cannot affect the perpendicular component. Hence its absolute length is the minimum distance.

## §1.3 A normal vector classifies the line–plane relation

A plane has equation

$$\Pi = \{x \in \mathbb{R}^3 : n \cdot (x - p_0) = 0\}, \qquad n \neq 0.$$

Substitute the line $x = p + tu$. Its intersection parameters satisfy

$$n \cdot (p - p_0) + t(n \cdot u) = 0.$$

Three cases are now explicit. If $n \cdot u \neq 0$, there is exactly one solution; the line crosses the plane. If $n \cdot u = 0$ but $n \cdot (p - p_0) \neq 0$, no solution exists, so the line is parallel to the plane. If both vanish, the entire line lies in the plane.

In particular, saying that the line's direction is perpendicular to the normal establishes only that it is *parallel to or contained in* the plane. It does not distinguish the two without a point test.

Two planes have parallel normal vectors if and only if their directions agree; the geometric planes are either coincident or disjoint parallel planes. If their normals are independent, the intersection is a line since the two linear equations have rank two.

## §1.4 Perpendicularity to two directions is enough

**Theorem 1.4.1**

Let $\Pi = p + \operatorname{span}\{u, v\}$, where $u, v$ are linearly independent. A direction vector $w$ is perpendicular to the plane if and only if

$$w \cdot u = w \cdot v = 0.$$

*Proof.* Every direction of $\Pi$ has the form $au + bv$. The assumptions imply

$$w \cdot (au + bv) = a(w \cdot u) + b(w \cdot v) = 0$$

for all scalars $a, b$. The converse follows by applying perpendicularity to the particular directions $u, v$. □

This is the algebraic content of the traditional test using two intersecting lines in a plane. The intersection requirement guarantees independent directions; two parallel lines would give redundant information.

## §1.5 Projection provides the closest point

Let $W$ be a linear subspace of a finite-dimensional inner-product space. Every $x$ decomposes uniquely as

$$x = p + r, \qquad p \in W, \quad r \in W^{\perp}.$$

For any $w \in W$, the Pythagorean theorem gives

$$\|x - w\|^2 = \|r + (p - w)\|^2 = \|r\|^2 + \|p - w\|^2 \geq \|r\|^2.$$

Thus $p$ is the unique closest point of $W$ to $x$. For an affine plane $p_0 + W$, project $x - p_0$ onto $W$ and then add $p_0$.

If $\Pi$ has normal $n$, the projection formula becomes

$$\operatorname{proj}_{\Pi}(x) = x - \frac{n \cdot (x - p_0)}{\|n\|^2}\, n,$$

and the distance from $x$ to the plane is

$$d(x, \Pi) = \frac{|n \cdot (x - p_0)|}{\|n\|}.$$

These are two views of the same orthogonal decomposition.

## §1.6 The three-perpendicular theorem as a dot product

Let $x \notin \Pi$, let $p = \operatorname{proj}_{\Pi}(x)$, and let $q \in \Pi$. The segment $x - q$ decomposes as

$$x - q = (x - p) + (p - q).$$

For any vector $v$ tangent to $\Pi$,

$$(x - q) \cdot v = (x - p) \cdot v + (p - q) \cdot v = (p - q) \cdot v.$$

Hence $xq$ is perpendicular in direction to a line of $\Pi$ with direction $v$ if and only if its projected segment $pq$ is perpendicular to that line. This proves the exact direction comparison underlying the three-perpendicular theorem, without relying on a potentially misleading sketch.

## §1.7 Incidence has an affine and a projective layer

The line $p + tu$ can be completed to a projective line, in which the direction $u$ determines a point at infinity. Two parallel affine lines share that projective point; two skew lines remain disjoint as projective lines in three-space. They do not automatically intersect merely because the space was projectively completed.

In $\mathbb{P}^3$, a projective plane is the zero set of a nonzero linear functional $\ell$ on $\mathbb{R}^4$, and incidence of the plane with a point $[X]$ is simply $\ell(X) = 0$. The dual language adds no metric: distance and perpendicularity still require an inner product, while intersection and parallel completion require only linear dependence.

The practical distinction is now precise. Rank tests decide incidence, normal vectors decide orientation relative to planes, dot products decide perpendicularity, and orthogonal decomposition computes shortest distances.